Dynamics: Problem Sheet 4 (of 8)

1. Consider the following system of coupled ODEs for $x(t)$, $y(t)$:

$$\begin{aligned} \ddot{x} &= 1 + \sin y - \mathrm{e}^{3x} \ , \\ \ddot{y} &= \mathrm{e}^{x-3y} - 1 \ . \end{aligned}$$

   (a) Show that $(x, y) = (0, 0)$ is an equilibrium configuration, and that the linearized equations of motion about this point are

$$\begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = M \begin{pmatrix} x \\ y \end{pmatrix} \ , \qquad \text{where } M = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix} \ .$$

   (b) By determining the eigenvalues and eigenvectors of M, hence show that the *normal mode solutions* to the equations in part (a) are

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = A \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos(\sqrt{2}t + \phi) \ , \qquad \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = B \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos(2t + \psi) \ ,$$

   where A, B, ϕ and ψ are constants.

2. Suppose that $(x(t), y(t))$ satisfy the following system of ODEs:

$$\begin{aligned} \ddot{x} &= 4x^3, \\ \ddot{y} &= 4y^3 \ . \end{aligned}$$

   (a) Show that the quantity

$$E = \frac{1}{2}\left(\dot{x}^2 + \dot{y}^2\right) + V(x, y),$$

   is constant, i.e. $\dot{E} = 0$, where

$$V(x, y) = -(x^4 + y^4).$$

   (b) A linear stability analysis about the equilibrium point $(0, 0)$ does not work (can you see why?). Argue, based on the shape of V, about the stability of this point.

3. A particle of mass m moves in $\mathbb{R}^3$ under the influence of a force $\mathbf{F} = -k\,\mathbf{r}$, where $\mathbf{r}$ is the position vector of the particle and $k > 0$ is constant.

   (a) Explain why $\mathbf{F}$ is both a *conservative* force, and a *central* force, where a choice of potential energy function is $V(\mathbf{r}) = \frac{1}{2}k|\mathbf{r}|^2$. Hence deduce that the particle moves in a plane through the origin.

   (b) Show that the general solution to the equation of motion is

$$\mathbf{r}(t) = \mathbf{A} \sin \omega t + \mathbf{B} \cos \omega t \ ,$$

   where $\mathbf{A}$ and $\mathbf{B}$ are constant vectors, and $\omega = \sqrt{k/m}$. (You might find it helpful to write out the vector equation of motion in terms of its components.)

   (c) Show that the solution in part (b) may be rewritten as

$$\mathbf{r}(t) = \mathbf{a} \sin(\omega t + \phi) + \mathbf{b} \cos(\omega t + \phi) \ ,$$

   where now $\mathbf{a}$ and $\mathbf{b}$ are constant *orthogonal* vectors, and ϕ is a constant phase.

(d) Hence show that the trajectory of the particle traces out an ellipse, with centre the origin.

(e) Taking the plane of motion to be the (x, y) plane, the solution to the equation of motion may be written as

$$\mathbf{r}(t) \quad = \quad a\,\sin(\omega t + \phi)\,\mathbf{i} + b\,\cos(\omega t + \phi)\,\mathbf{j}\ ,$$

where $\omega = \sqrt{k/m}$, and a, b and ϕ are constant. Compute the total energy E and total angular momentum $\mathbf{L}$ about the origin, thus confirming that both are indeed constant. Show in particular that the *specific angular momentum* $|\mathbf{L}|/m = 2A/T$, where A is the area of the ellipse traced out by the solution, and T is the period of the solution.

4. At a given instant of time, a particle of mass m has position vector $\mathbf{r}$, measured from the origin O of an inertial frame, and velocity $\mathbf{v}$. Let $\mathcal{L}$ be the straight line through $\mathbf{r}$ with tangent vector $\mathbf{v}$. Show that the angular momentum $\mathbf{L}_O$ of the particle about O has magnitude $|\mathbf{L}_O| = d|\mathbf{p}|$, where d is the perpendicular distance between O and $\mathcal{L}$, and $\mathbf{p}$ is the (linear) momentum of the particle. When is $\mathbf{L}_O = \mathbf{0}$?